

Muhammad Asfandyar

AI & Machine Learning | Full Stack | Cybersecurity

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Professional Summary

Motivated Computer Science undergraduate (7th semester) with a strong command of full-stack web development and solid hands-on skills in AI, Machine Learning, and Cybersecurity. Has designed, built, and deployed multiple live projects — full-stack e-commerce platforms, AI-powered applications, and data-driven dashboards — from concept through to production. Certified Ethical Hacker (CEH) with a security-first mindset. A quick learner and team player eager to grow under senior engineers and contribute to junior developer, AI/ML, or security roles, delivering clean, reliable results on schedule.

Core Skills

Full-Stack Web Development

- Strong command of web development — responsive, production-ready websites & web applications
- MERN stack — React (Vite), Node.js, Express & MongoDB
- Next.js (App Router) & TypeScript
- REST API design, CRUD operations & integration
- Authentication & security — JWT, bcrypt, protected routes
- HTML5, CSS3, JavaScript, Tailwind CSS & modern front-end frameworks
- Converting designs into polished, cross-browser, mobile-friendly pages

AI & Machine Learning

- Deep learning — TensorFlow / Keras, transfer learning (MobileNetV2)
- Image classification — crop disease detection (38 classes, 92–96% accuracy)
- Model explainability — Grad-CAM heatmaps for transparent, trustworthy AI
- Image processing — OpenCV, Pillow; dataset handling (PlantVillage, 54K+ images)
- LLM integration — Google Gemini 2.5 Flash for natural-language querying & smart CSV analysis
- AI-powered product features — insights, recommendations & automated reporting
- Built and deployed multiple AI/ML projects as live web applications (Streamlit)

Cybersecurity

- Certified Ethical Hacker (CEH) — EC-Council
- Hands-on CEH v13 labs — footprinting & reconnaissance, network scanning, enumeration, vulnerability analysis, system hacking & malware threats
- Practical tooling — Nmap, Burp Suite & Metasploit
- Web application security — OWASP Top 10
- Security-first approach to development, hardening & code review

Tools & Platforms

- Python, TensorFlow, OpenCV, Streamlit, Google Gemini
- JavaScript, Next.js, React, Node.js
- Git & GitHub, VS Code, Vercel, cPanel
- Deployment, domains & hosting management
- Windows & Linux environments

Experience

Self-Employed Web Developer | Freelance

- Designed and built responsive, mobile-friendly websites and web applications for clients, converting requirements and designs into clean, cross-browser compatible pages.
- Developed and customized full-stack solutions end-to-end — structure, styling, functionality, and deployment to live hosting (Vercel, cPanel).
- Independently handled hosting, domains, and deployment for all live projects (Vercel, Streamlit, cPanel), from build through to going live in production.
- Deployed and maintained live production projects, handling updates, fixes, and performance.
- Troubleshoot website errors, layout issues, and technical problems, ensuring sites ran smoothly after launch.

Projects

OpsPulse — AI-Powered Business Command Center

<https://v0-opspulse.vercel.app/>

- Built a real-time operational intelligence platform that turns raw business data into actionable insights using AI.
- Implemented three Google Gemini 2.5 Flash features — natural-language querying with instant charts and insights, smart CSV analysis (summaries, patterns, anomaly detection), and automated executive report generation.
- Real-time dashboards with live KPI cards and advanced analytics (Recharts), anomaly detection with severity-level alerts, and dark/light mode in a responsive UI. Built with Next.js 16, TypeScript, Tailwind CSS v4, Radix UI & PapaParse, with API routes and intelligent fallbacks.

CropGuard AI — Crop Disease Detection with Grad-CAM

<https://cropguardai-project.streamlit.app/>

- AI crop disease detection — a farmer uploads a leaf photo and the app detects the disease across 38 disease classes with a confidence level, estimates severity (Mild / Moderate / Severe), and recommends a treatment.
- Built a MobileNetV2 transfer-learning model (TensorFlow/Keras) trained on the PlantVillage dataset (54,000+ images, 38 classes) with two-phase training (frozen → fine-tune), achieving 92-96% validation accuracy.
- Implemented Grad-CAM explainability — a heatmap overlay showing exactly where on the leaf the disease was detected (OpenCV, Pillow) — demoed via a live Streamlit web app.

GemsInsider — Natural Gemstones E-Commerce Store

<https://gemsinsider.co>

- Built and deployed a full-stack MERN e-commerce website for a natural gemstones brand — React front end (Vite) with a Node.js/Express REST API and MongoDB database.
- Implemented user registration and login with JWT authentication, shopping cart, checkout and payment flow, order management, and an admin panel for managing orders.
- Product catalog with categories, global search and color filtering, image galleries, contact form, light/dark theme, and policy / FAQ pages — fully responsive across devices.

Intrusion Detection System — Machine Learning & Flask

- Built a web-based Intrusion Detection System that simulates network attack scenarios (DoS, DDoS, port scans) and detects attacks in real time from network traffic data.
- Trained and compared four scikit-learn models — Decision Tree, K-Nearest Neighbors, Logistic Regression, and Random Forest — showing predictions, confidence, and accuracy side by side.
- Flask web interface with pandas-based data handling; supports uploading custom network traffic for analysis, with visualizations of the network features driving each prediction.

AuraTickets — Ticket Booking System

- Built a full ticket booking system with PHP 8, MySQL, and Bootstrap 5 on XAMPP — landing page, live booking form, booking-confirmation page with ticket summary, and an admin dashboard to add events and inspect bookings.
- Designed a relational database with foreign keys linking events and bookings.

Medical LMS — Learning Management System (Team of 2)

- Built a full-stack MERN learning management system in a 2-member team (with Zaka Ullah) — React 19 front end (Vite, Tailwind CSS 4, framer-motion, react-router) with an Express 5 and MongoDB back end.
- Implemented JWT authentication, rich-text content authoring (Quill), form validation (react-hook-form), and Excel-based data import (xlsx).
- Structured the backend with MVC (routes, controllers, models, middleware) plus migrations and seeders; deployed on Vercel.

Education & Certifications

BS Computer Science

University of Engineering and Technology, Peshawar

7th semester | Expected graduation 2027

Intermediate (FSc)

Cadet College Razmak

Certified Ethical Hacker (CEH)

EC-Council

Languages

Urdu: Native English: Professional Working Proficiency